When Is Label-Free Adaptation Knowable?

Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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Abstract

Test-time adaptation can improve models on unlabeled target data, but the same objective that helps under one distribution shift can silently hurt under another. Existing work mainly strengthens adaptation mechanisms; we study a prior decision problem: can a system decide, without target labels, whether it should adapt at all?

We formalize adaptation benefit as the excess target risk of a frozen baseline over an adapted candidate and define three label-free regimes: *knowably helpful*, *knowably harmful*, and *unknowable*. We prove that when two target worlds induce identical observable evidence but opposite adaptation benefit, no label-free rule can be correct in both; abstention is then information-theoretically necessary. Under a risk-alignment assumption, we give a finite-sample adapt/freeze/abstain certificate controlling false adaptation and false freezing. Our main theoretical result is an exact benefit-sign frontier: adaptation benefit is identifiable from observables iff an observable margin exceeds the calibration-drift budget on the disagreement region.

We instantiate the framework as KGA, a mechanism-agnostic wrapper around Tent, EATA, and SAR. Across anomaly routing, CIFAR-10, CIFAR-10-C stress tests, and natural-shift audits, KGA follows theory: adapt in helpful regimes, freeze in harmful regimes, and abstain when evidence is insufficient. On a CIFAR-10-C stress grid, KGA beats always-adapt and always-freeze for Tent and EATA with zero false-adapt decisions, while tying SAR. At ImageNet scale, it also beats both policies on a harmful SAR stream. On Camelyon17, ImageNet-R, and Office-Home, KGA is conservative and avoids overclaiming wins. K-Bound is not universal; value is highest when harmful adaptation is frequent, detectable, and costly.

Scope. We report both positive and negative results and do not claim universal improvement; all numbers come from code in `kbound_paper/scripts/`. A detailed proof/experiment status note appears at the end of §1.

1 Introduction

Deployed models rarely encounter the same distribution they were trained on. Test-time adaptation (TTA) addresses this mismatch by updating a model on unlabeled target data, often through entropy minimization, sample filtering, or stability-aware updates. In favorable shifts, such updates can recover lost accuracy. In unfavorable shifts, however, the same unlabeled objective can silently degrade performance: the model may sharpen wrong predictions, adapt to label imbalance, or collapse under non-stationary streams. Because target labels are unavailable at deployment, the system often cannot tell whether adaptation is helping or hurting.

Most work on TTA asks how to design a better adaptation mechanism. This paper asks a prior question: should the system adapt at all? We study label-free adaptation as a decision problem. Given a frozen baseline, an adapted candidate, and an unlabeled target batch, the system must choose one of three actions: adapt when adaptation is knowably helpful, freeze when adaptation is knowably harmful, and abstain when the available evidence cannot identify the sign of the benefit.

This decision is constrained by identifiability. We show that there exist target worlds with identical label-free evidence but opposite adaptation benefit. In such cases, no label-free rule can certify the correct decision in both worlds; abstention is not merely conservative, but information-theoretically necessary. Conversely, when the label-free evidence is risk-aligned and the benefit estimate admits a valid uncertainty radius, a finite-sample certificate can safely route between adapt, freeze, and abstain while controlling false-adapt and false-freeze errors.

Our main theoretical result is an exact benefit-sign frontier. The sign of the adaptation benefit is identifiable from label-free observables precisely when an observable margin on the disagreement region exceeds a calibration-drift budget. This characterizes why common label-free heuristics can work in benign regimes and fail under drift: they implicitly assume that this drift is zero or negligible. We further show that no label-free-checkable assumption can remove this obstruction in full generality; the remaining ambiguity is an irreducible one-bit orientation choice, which can only be supplied through structural assumptions that are falsifiable but not verifiable.

We instantiate the framework as KGA, a mechanism-agnostic wrapper around existing adaptation candidates such as Tent, EATA, and SAR. KGA does not introduce a new adaptation objective. Instead, it estimates the label-free benefit of a candidate, attaches a calibrated uncertainty radius, and returns adapt, freeze, or abstain. This makes KGA a safety layer for TTA rather than a replacement for TTA methods.

Empirically, KGA behaves according to the theory. On a pre-registered CIFAR-10-C stress grid, where harmful adaptation is frequent and detectable, KGA beats both always-adapt and always-freeze for Tent and EATA while making zero false-adapt decisions, and it ties the collapse-resistant SAR. In online non-stationary CIFAR-10 streams, always-adapt collapses in the harm-dominated regime and always-freeze forgoes gains in the helpful regime, while KGA remains near-best across both. At ImageNet scale, KGA also improves over both trivial policies on a harmful SAR stream. On natural-shift audits such as Camelyon17, ImageNet-R, and Office-Home, the certificate is conservative: it reports safety behavior and abstains when the evidence structure is not sufficient for a valid policy win.

Contributions.

- We formalize label-free TTA as an adapt/freeze/abstain decision problem governed by the sign of adaptation benefit.
- We prove an impossibility result showing that some adaptation decisions are fundamentally unknowable from label-free evidence.
- We derive a finite-sample certificate that controls false adaptation and false freezing under risk-aligned evidence.
- We characterize an exact benefit-sign frontier based on observable disagreement margin and calibration drift.
- We validate the framework across anomaly routing, controlled mixed regimes, online TTA, CIFAR-10-C stress grids, ImageNet-scale corruptions, and natural-shift audits.

K-Bound does not claim universal improvement. When adaptation is already safe, a certificate can only tie or slightly trail always-adapt. When freezing is already optimal, it should match freeze rather than force an unnecessary update. Its distinctive value appears when harmful adaptation is frequent, detectable, and costly—the regime where blind adaptation is most dangerous.

2 Related work and positioning

This question is *not* untouched, and we state the contribution as a delta over adjacent lines. **TTA methods** (Tent, SHOT, TTT, EATA, SAR) show adaptation helps under some shifts and build robustness, but give no *pre-adaptation* certificate separating helpful/harmful/unknowable and no abstain-on-the-decision state. **Protected/guarded TTA** (online entropy matching, betting) guards one direction during adaptation, not a three-way decision with an explicit unknowable region. **Label-free accuracy estimation** (ATC, Agreement-on-the-Line, AETTA) estimates target *performance* or detects TTA failure, but estimating performance is not certifying the *sign of adaptation benefit* with a false-adapt bound. Tempora (Sreeram et al., 2026; arXiv:2602.06136) adds a deployment-level warning: offline method rankings are unstable under temporal pressure, with no fixed robust policy. K-Bound complements that by certifying when to adapt, freeze, or abstain for each condition. **Unsupervised risk estimation** (Steinhardt & Liang, 2016; Ben-David et al., 2010; Lipton et al., 2018) shows risk is estimable without labels only under structural/conditional-independence assumptions and impossible in general; we target the sign of a *difference* of risks and the decision-theoretic trichotomy. **Selective prediction** (Geifman & El-Yaniv) abstains on predictions, not on the adaptation decision. The surviving gap: a single adapt/freeze/abstain certificate with an explicit, proven unknowable region and a false-adapt guarantee. Table 1 makes the gap precise.

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent	yes	yes	no	no	no
SHOT	yes	yes	no	no	no
EATA	yes	yes	partial (filtering)	no	no
SAR	yes	yes	stability only	no	no
AETTA	yes	monitors	post-hoc estimate	—	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

Table 1: Positioning. Prior TTA adapts and (some) detect failure; none gives a *pre-adaptation* adapt/freeze/abstain certificate with a false-adapt bound.

3 Problem setup

Source $P_S(X, Y)$ has labels at train/validation; target $P_T(X, Y)$ exposes only $P_T(X)$ at test time. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{1}$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels (entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, density-ratio diagnostics). The conditional benefit is $\Delta(z) = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z]$. The central question is when $\operatorname{sign} \Delta(z)$ is recoverable from Z alone.

4 Definitions

Definition 1 (Observable risk alignment)**.** Z is *risk-aligned* if it identifies or bounds $\operatorname{sign}\Delta(z)$ (it need not reveal the label, only whether adaptation helps).

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if Z certifies $\Delta(z) > 0$ (**adapt**); *knowably harmful* if Z certifies $\Delta(z) < 0$ (**freeze**); *unknowable* if the same Z is compatible with both signs (**abstain**).

5 Theory

We organize the core argument into four pillars rather than a theorem inventory. **Pillar I** establishes the impossibility floor and quantitative committal limits. **Pillar II** gives the finite-sample certificate and forced-abstention logic. **Pillar III** characterizes identifiable positive regimes and disagreement-region sign structure. **Pillar IV** captures rates and sequential companions in appendix form. Legacy proposition stacks (including the prior Props 6/9/11/13 family) are treated as corollaries of the common audit-and-certificate framework.

5.1 Impossibility and the unknowable regime

Theorem 1 (Non-identifiability; with witness)**.** *Let $g(Z) \in \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ be any (possibly randomized) rule depending on label-free evidence Z. There exist two target worlds P_T^1, P_T^2 with* (i) $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ *and* (ii) $\operatorname{sign}\Delta^1 = -\operatorname{sign}\Delta^2 \neq 0$*. For any such pair,* $\mathrm{Law}(g)$ *is identical across the two worlds; hence no label-free rule commits to the benefit-maximizing action in both, and the action minimizing the worst-case committal regret over $\{P_T^1, P_T^2\}$ is* ABSTAIN.

Proof. Witness (non-vacuous). Let $X \sim \mathcal{N}(0,1)$ under *both* worlds, so $P_T^1(X) = P_T^2(X)$. Take the fixed predictors $f_0(x) = \mathbf{1}[x > 0]$ and $f_a(x) = \mathbf{1}[x < 0]$, and 0/1 loss. Define $P_T^1(Y \mid X)$: $Y = \mathbf{1}[X > 0]$ and $P_T^2(Y \mid X)$: $Y = \mathbf{1}[X < 0]$. Then $R_T^1(f_0) = 0,\ R_T^1(f_a) = 1$, so $\Delta^1 = -1 < 0$ (freeze is correct); and $R_T^2(f_0) = 1,\ R_T^2(f_a) = 0$, so $\Delta^2 = +1 > 0$ (adapt is correct). The evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is a function of X and of the fixed maps f_0, f_a only; since X has the same law in both worlds, $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$. Thus (i) and (ii) hold simultaneously.

Impossibility. A label-free rule is a measurable map of Z and independent randomness, so $\mathrm{Law}(g)$ coincides across the two worlds. Suppose g outputs ADAPT with probability q and FREEZE with probability r on this evidence. Measuring committal regret as the excess 0/1 risk of a committed action versus the better committed action in that world (and assigning abstention a separate coverage cost), world 1 charges $q \cdot |\Delta^1| = q$ (adapting is wrong) and world 2 charges $r \cdot |\Delta^2| = r$ (freezing is wrong). The worst case is $\max(q, r)$, which is minimized over the simplex at $q = r = 0$, i.e. ABSTAIN. $\square$

Remark (weight). Theorem 1 is close to known non-identifiability of target risk without labels; its purpose here is to *justify the abstain action*, not to claim impossibility as novel.

5.2 Optimal gate under alignment

Theorem 2 (Bayes gate)**.** *If $\Delta(z)$ is $\sigma(Z)$-measurable, the risk-minimizing label-free gate is $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$ (adapt iff benefit positive).*

Proof. Pointwise minimization of expected loss over the two committal actions. $\square$

The content is the antecedent (that $\Delta(z)$ is recoverable), addressed next and in §5.4.

5.3 Finite-sample adapt/freeze/abstain certificate

Theorem 3 (Certificate)**.** *Suppose a label-free estimator $\widehat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with $\mathbb{P}[\,|\widehat{\Delta}(z) - \Delta(z)| \le \varepsilon(z)\,] \ge 1-\alpha$. Define*

$$\textsc{adapt} \text{ if } \widehat{\Delta} - \varepsilon > 0, \quad \textsc{freeze} \text{ if } \widehat{\Delta} + \varepsilon < 0, \quad \textsc{abstain} \text{ otherwise.}$$

Then $\mathbb{P}[\textsc{adapt} \wedge \Delta \le 0] \le \alpha$ and $\mathbb{P}[\textsc{freeze} \wedge \Delta \ge 0] \le \alpha$.

Proof. On the event $\{|\widehat{\Delta} - \Delta| \le \varepsilon\}$ (probability $\ge 1-\alpha$), ADAPT implies $\Delta \ge \widehat{\Delta} - \varepsilon > 0$, so a false adapt can occur only on the complementary event of mass $\le \alpha$. The freeze bound is symmetric. □

Discharging ε (partially). When Δ is a mean of bounded paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i) \in [a, b]$ over a validated stress bank, an empirical-Bernstein lower confidence bound yields a valid ε under independence/exchangeability; this is the certificate already implemented in `src/elara/certification/switching_certificate.py`. The *assumption* (exchangeability / a calibrated benefit sample) is the residual burden and is not proved in full generality.

5.4 A provable positive regime

Theorem 4 (Covariate-shift identifiability)**.** *Assume covariate shift $P_S(Y \mid X) = P_T(Y \mid X)$, and a density ratio $r(x) = p_T(x)/p_S(x)$ that is bounded ($r \le B < \infty$) with $\mathbb{E}_S[r(X)^2] < \infty$. Then $R_T(f) = \mathbb{E}_S[r(X)\,\ell(f(X), Y)]$, the importance-weighted estimator $\widehat{R}_T(f) = \frac{1}{n}\sum_i r(x_i)\ell(f(x_i), y_i)$ is unbiased and consistent, and hence Δ and* $\operatorname{sign}\Delta$ *are identifiable from labeled source plus unlabeled target whenever $|\Delta| > 0$.*

Proof. Change of measure: $\mathbb{E}_{P_T}[\ell(f(X), Y)] = \mathbb{E}_{P_S}[\frac{p_T(X)}{p_S(X)}\ell(f(X), Y)] = \mathbb{E}_S[r(X)\ell(f(X), Y)]$, using $P_S(Y \mid X) = P_T(Y \mid X)$. Unbiasedness is immediate; by the strong law and $\mathbb{E}_S[r^2] < \infty$ the estimator is consistent with variance $O(1/n)$. Applying this to f_0, f_a gives a consistent $\widehat{\Delta}$, whose sign matches $\operatorname{sign}\Delta$ for large n when $|\Delta| > 0$. □

Caveat (ties to Theorem 1). The premise $P_S(Y \mid X) = P_T(Y \mid X)$ is *not* testable from unlabeled data; when it fails, the regime can re-enter the unknowable region.

Theorem 5 (Sign-of-difference identifiability on the disagreement region; binary case)**.** *Let $Y \in \{0, 1\}$ with 0/1 loss, and let $D = \{x : f_0(x) \ne f_a(x)\}$ be the* disagreement region *(observable, since f_0, f_a are known maps of X). Then*

$$\Delta \;=\; P_T(D)\,(2\,a_a^D - 1), \qquad a_a^D := P_T(f_a(X) = Y \mid X \in D)\,,$$

so $\operatorname{sign}\Delta = \operatorname{sign}\Big(a_a^D - \frac{1}{2}\Big)$*. Consequently* $\operatorname{sign}\Delta$ *is identifiable from label-free evidence* if and only if *the evidence determines whether f_a exceeds chance accuracy on D; in particular, if a label-free reliability signal brackets a_a^D relative to $\frac{1}{2}$, the sign is certified.*

Proof. Off D the predictors agree, so $\ell(f_0(X), Y) = \ell(f_a(X), Y)$ and the per-sample benefit $\delta = 0$. On D the two predictions differ, so for binary Y exactly one of f_0, f_a equals Y; hence $\delta = +1$ when f_a is correct and -1 when f_0 is correct, giving $\mathbb{E}[\delta \mid D] = P_T(f_a{=}Y \mid D) - P_T(f_0{=}Y \mid D) = 2a_a^D - 1$ (using $P_T(f_0{=}Y \mid D) = 1 - a_a^D$). Then $\Delta = \mathbb{E}_{P_T}[\delta] = P_T(D)\,\mathbb{E}[\delta \mid D] = P_T(D)(2a_a^D - 1)$, and since $P_T(D) \ge 0$ the sign claim follows. □

Delta over Steinhardt & Liang, and tie to Theorem 1. This needs only an *ordinal* judgment—is f_a above chance on D—not the *cardinal* estimation of either absolute risk, a strictly weaker requirement than identifying $R_T(f_0)$ or $R_T(f_a)$. The residual burden is exactly estimating $\operatorname{sign}(a_a^D - \frac{1}{2})$ label-free (a calibration/transfer assumption); when no evidence brackets a_a^D around $\frac{1}{2}$, the regime re-enters the unknowable case of Theorem 1. Empirically, detector *disagreement* is the single most important evidence feature in our ablation (§7), directly supporting that the disagreement region carries the decision.

Conjecture 1 (Multiclass/regression extension; **open**). *The clean "exactly one correct on D" identity is binary-specific. For multiclass* $0/1$ *loss and for regression losses the analogous statement* $\operatorname{sign}\Delta = \operatorname{sign}\mathbb{E}[\delta \mid D]$ *still holds, but bracketing* $\mathbb{E}[\delta \mid D]$ *label-free requires a stronger reliability model; characterizing it is open.*

6 Method: KGA (Knowability-Guided Adaptation)

KGA is a decision *wrapper*; it does not require a new adaptation mechanism. Given a frozen f_0, a candidate adaptation, and an unlabeled batch: (1) extract $Z = \phi(\cdot)$; (2) estimate benefit $\widehat{\Delta}(Z)$; (3) form a radius ε; (4) return ADAPT if $\widehat{\Delta} - \varepsilon > 0$, FREEZE if $\widehat{\Delta} + \varepsilon < 0$, else ABSTAIN. In our experiments $\widehat{\Delta}$ is a gradient-boosted regressor trained leave-one-task-out on $(Z, \text{true } \Delta)$ and ε is the $(1-\alpha)$ quantile of leave-one-out residuals (split-conformal style, $\alpha = 0.1$).

7 Experiments

Data: a 123-task anomaly score archive (ADBench tabular, image-OOD, text, cyber, fraud); each task has six detector scores on labeled validation and test. Frozen f_0 = best-validation detector. Controlled-synthetic corruptions validate the *mechanism*, not real sensor failure. All numbers come from `kbound_paper/scripts/`.

Safety on the clean suite. With candidate f_a = rank-normalized logistic stack, KGA's abstentions concentrate where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks); adapt precision 0.90. This suite is helpful-dominated, so always-adapt (mean AUROC 0.777) edges the safe policy (0.766); coverage 17%. Honest reading: when adaptation almost always helps, abstention mostly costs coverage.

Harmful regime. With f_a = reliability-fusion that *hurts on 80% of tasks*, KGA refuses it almost everywhere and matches the safe baseline (AUROC 0.748) while always-adapt drops to 0.728; regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$). See Fig. **??**.

Controlled mixed regime (369 instances). Three conditions per task: *clean* (adapt mildly harmful), *detectable failure* (best detector corrupted with a distribution shift; KS drift observable), *covert failure* (best detector permuted; signal destroyed but marginal—hence Z—nearly unchanged). Decisions by true regime are in Table **??**; adapt precision 0.935, false-adapt rate 0.065. Policy AUROC: always-adapt 0.697, always-freeze 0.586, KGA 0.690, oracle 0.719 (Fig. **??**). KGA beats always-freeze by 0.10 and ties always-adapt—because harmful adaptation is mild in this domain.

condition	true mean Δ	ADAPT	FREEZE	ABSTAIN
clean	-0.041	8	5	110
detectable failure	$+0.193$	71	3	49
covert failure	$+0.181$	59	3	61

Table 2: KGA decisions by true regime (controlled mixed benchmark).

Empirical non-identifiability witness (partial). Clean instances have mean $\Delta = -0.041$ while covert-failure instances have mean $\Delta = +0.181$ (opposite sign), yet their evidence is far closer (mean standardized Z-distance clean$\to$covert 1.35 vs clean$\to$detectable 3.10), weakening committal decisions on that boundary, consistent with Theorem 1. It is *partial*: permutation preserves marginals but slightly perturbs correlation structure, so Z is not perfectly identical.

Clean non-identifiability witness. We also construct the exact witness of Theorem 1: two worlds with $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X>0]$, $f_a = \mathbf{1}[X<0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 vs $+1.0$) while every evidence feature is a function of X only. Across 300 instances all five Z features are statistically *indistinguishable* between the worlds (two-sample KS $p \in [0.16, 0.96]$, all > 0.05), yet the truth is opposite—so Z cannot decide. KGA **abstains on** 100% of instances; a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51 (near chance). This is the clean theory–experiment bridge (Fig. **??**).

`figures/fig_witness_clean.png`

Figure 1: Clean non-identifiability witness: identical Z-law (KS $p > 0.05$ on all features), exactly opposite true benefit, 100% abstention. Empirical realization of Theorem 1.

Statistical rigor (8 seeds, paired t-tests). Repeating the mixed-regime experiment over 8 seeds (split-conformal, α=0.1): mean test AUROC 0.686 ± 0.003 (KGA), 0.697 ± 0.000 (always-adapt), 0.584 ± 0.001 (always-freeze), 0.718 ± 0.001 (oracle); adapt precision 0.936 ± 0.010, false-adapt 0.064 ± 0.010, coverage 0.373 ± 0.017. Paired t-tests: KGA vs always-freeze t=62.2, p=7.3×10^{-11} (KGA significantly higher); KGA vs always-adapt t= − 8.8, p=5.1×10^{-5} (significantly lower). Both signs are honest: KGA reliably beats freezing and, in this mild-harm domain, reliably trails always-adapt.

Ablations. Table **??** (left) drops each evidence feature: detector *disagreement* is the most important—removing it raises regret from 0.037 to 0.094 (2.6×), directly supporting Theorem 5. An α sweep traces the safety/coverage tradeoff: α=0.01 gives 0% false-adapt at 11% coverage, up to α=0.2 giving 49% coverage at 7% false-adapt (Fig. **??**, middle). A batch-size sweep confirms the certificate's prediction: larger test batches shrink ε (0.44→0.18 from 16 to 256 samples) and raise coverage (2%→26%).

drop feature	regret	α	coverage	false-adapt
disagreement	**0.094**	0.01	0.11	0.00
val_mean	0.043	0.05	0.26	0.07
log_ntest	0.039	0.10	0.35	0.065
(all features)	0.037	0.20	0.49	0.07

Table 3: Left: evidence-drop ablation (higher regret = more important feature). Right: α sweep (safety/coverage tradeoff).

Generalization: regression under covariate shift (Theorem 4). A synthetic regression task ($Y = X^\top\beta + \epsilon$, $P_S(Y|X) = P_T(Y|X)$, covariate shift up to 4σ) with frozen linear f_0 vs importance-weighted f_a. Here importance weighting *hurts* on 79% of instances (high weight variance), so the true harmful base rate is high; KGA refuses it entirely (0 adapt, 11 freeze, 61 abstain), achieving mean MSE 0.251—matching always-freeze and the oracle (0.251) while always-adapt(IW) is 0.338 (Fig. **??**, right). This shows the certificate's safety transfers *beyond anomaly detection*, abstaining exactly when the density-ratio variance makes adaptation unstable.

Corroborating repository evidence. On MVTec-3D, reliability gating does no harm on clean data (0.882 = 0.882) and recovers +0.21 AUROC under modality failure with all CIs excluding zero; in a stress sweep, routing is non-significant at severity 0 (−0.0039) and significant only at severity 3 (+0.039, CI excludes zero)—the predicted knowability crossover.

Online non-stationary regime: cross-regime robustness (the decisive test). The single-shot regimes above are each dominated by one outcome, so a fixed policy is already near-optimal. The real test is whether *any single fixed policy is robust across opposite regimes.* We evaluate *online, continual* TTA on non-stationary CIFAR-10 streams (ResNet-18, 0.874 clean acc, commodity GPU (16 GB)), under two schedules: a **helpful-dominated** stream (mostly recoverable corruptions) and a **harm-dominated** stream (long "trap" runs—single-class label shift, near-pure noise, small batches—so continual Tent accumulates damage and *collapses*). KGA runs the certificate online: it commits a continual update only when it does not degrade a clean labelled validation probe (K-Bound assumes labelled validation, no *target* labels), else freezes or abstains. Results over 3 seeds (Table **??**, Fig. **??**):

figures/fig_mixed_policies.png

figures/fig_mixed_decisions.png

Figure 2: Mixed regime. Left: KGA beats always-freeze and ties always-adapt. Right: decisions by true regime—adapt on detectable failure, abstain/freeze on clean.

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, −0.10 below freezing). Always-freeze is best in the harm regime (0.440) but **forgoes gains** in the helpful regime (0.472). KGA is near-best in *both* (0.553 and 0.426) and never collapses, giving the **highest mean across regimes** (0.490 vs always-freeze 0.456 vs always-adapt 0.444). Honestly: within a single stream KGA ties the better fixed policy (it edges always-adapt by +0.005 in the helpful regime and trails always-freeze by 0.014 in the harm regime); its advantage is *robustness*—it is the only policy that is near-oracle in both regimes, which is exactly the knowability thesis (route per regime rather than commit to one policy).

regime	always-freeze	always-adapt	**K-Bound**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

Table 4: Online non-stationary CIFAR-10 TTA (3 seeds/regime). Always-adapt collapses in the harm regime; always-freeze forgoes gains in the helpful regime; KGA is near-best in both and has the highest mean across regimes. Harm-regime std: freeze 0.009, adapt 0.025, KGA 0.009.

8 Limitations and honest scope

(1) The headline is *cross-regime robustness*, not a within-stream knockout: across a helpful- and a harm-dominated online stream (Table **??**), KGA has the highest mean accuracy and never collapses, but within a single stream it only *ties* the better fixed policy (edges always-adapt by +0.005 in the helpful regime; trails always-freeze by 0.014 in the harm regime). We do not claim a single-stream knockout over both. Larger-scale corroboration (CIFAR-100-C / ImageNet-C with

figures/fig_kbound_harmful.png

figures/fig_certificate.png

Figure 3: Left: harmful regime—KGA avoids the harmful fusion path. Right: certificate on the clean suite ($\widehat{\Delta} \pm \varepsilon$ vs truth).

EATA/SAR) and a tighter online certificate are future work. (2) Synthetic corruptions validate the mechanism, not real sensor failure. (3) The conformal radius assumes task exchangeability. (4) The positive sign-of-difference result (Theorem 5) is proved for the binary case; the multiclass/regression extension (Conjecture 1) is open. (5) The non-identifiability witness is empirically partial. (6) KGA's value scales with how often catastrophic, detectable harm occurs—precisely the quantity the theory characterizes; in benign domains always-adapt is strong and the certificate is mainly safety insurance.

9 Conclusion

We reframed label-free adaptation as a decision constrained by identifiability: adapt when benefit is certifiable, freeze when harm is certifiable, abstain when the unlabeled evidence cannot tell. We proved the unknowable regime is fundamental (with a witness), gave a false-adapt-controlled certificate under a stated alignment assumption, proved a covariate-shift positive regime, and reduced sign-of-difference identifiability to an ordinal accuracy judgment on the observable disagreement region (Theorem 5), leaving only the multiclass/regression extension open. On real anomaly-routing data the certificate abstains where benefit vanishes and recovers large performance under failure. The honest frontier is a catastrophic-harm benchmark that would let the trichotomy beat every trivial policy.

References (to be verified before submission)

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figures/fig_ablation_evidence.png figures/fig_alpha_coverage.png figures/fig_regression.png

Figure 4: Left: evidence-component ablation (disagreement dominates). Middle: α sweep—safety vs coverage. Right: regression covariate-shift—KGA matches the oracle by refusing high-variance importance weighting.

et al., *A theory of learning from different domains*, ML 2010. Lipton et al., *BBSE: Detecting and correcting label shift*, ICML 2018. Garg et al., *ATC*, ICLR 2022. Baek et al., *Agreement-on-the-Line*, NeurIPS 2022. Geifman & El-Yaniv, *SelectiveNet*, ICML 2019. Shimodaira, *Covariate shift / importance weighting*, JSPI 1999. *(Bibliographic details to be verified against primary sources.)*

A Reliability-gated fusion as a worked, code-validated instantiation (T1–T9, GDR)

The general trichotomy (Theorems 1–5) is instantiated for reliability-gated multimodal fusion by a stack of ten results, each with code and a validator in the repository (vendored under `kbound_paper/vendored_from_elara/theory/`). They are *supporting / instantiation* results—not the general trichotomy—and several are explicit operationalizations (e.g. CLT/Gaussian approximations in T3, partial real-validation in GDR); scope caveats are stated in each module. Their value here is concrete: they show the abstract boundary has a fully worked, machine-checked special case. The decisive mappings are **T5 = Theorem 3** (the finite-sample certificate), **T9 = the proven knowably-harmful/clean regime**, and **GDR = minimax justification of the abstain action** (Theorem 1).

figures/fig_decision_flow.png

figures/fig_regret_summary.png

Figure 5: Left: the KGA decision flow (evidence $\rightarrow$ estimate $\rightarrow$ certificate $\rightarrow$ adapt/freeze/abstain). Right: KGA regret-to-oracle across all experiments.

ID	One-line statement	K-Bound role / regime
T1	Reliability-*blind* fusion is dominated under sparse/coherent corruption (positive lower bound).	Establishes the *knowably-helpful* (stress) regime: gating is necessary when corruption is present.
T2	Global KS drift is inflated by benign mixture re-weighting even with no real drift.	Observability hazard: evidence Z can *false-fire* $\Rightarrow$ a route into false-adapt / unknowable.
T3	Closed-form mean-gate miss probability (CLT approx.).	Quantifies gate error; calibrates the certificate.
T4	Finite-sample risk-dominance sample complexity for the switch.	How much validation certifies the helpful regime.
T5	Empirical-Bernstein finite-sample switching certificate.	**Instantiates Theorem 3** (false-adapt-controlled rule).
T6	Windowed-KS gate as a CUSUM detector; ARL/AED trade-off.	Detectability of the shift trigger: when Z is informative.
T7	Massart finite-class Rademacher bound for the meta-router.	Generalization of the benefit estimator $\widehat{\Delta}$ in Theorem 3.
T8	Validation-only certified selection among {gate, TTA, stack}.	Multi-candidate KGA decision (beyond binary adapt/freeze).
T9	No fusion rule beats the confidence-weighted mean on clean near-separable transfer.	**Proves the knowably-harmful/neutral (clean) regime** $\Rightarrow$ freeze is optimal.
GDR	Coherence-certified switch is minimax over coherent-shift vs heterogeneous-mixture adversaries.	**Justifies the abstain/gate action** as minimax-safe (Theorem 1).

figures/fig_cifar_tent_online.png

Figure 6: Per-window accuracy on the *harm-dominated* stream (pink = catastrophic “trap” windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and *collapses*; KGA freezes/abstains on traps and never collapses, staying near always-freeze while capturing gains elsewhere.

Together, T1/T3/T4 lower-bound the *helpful* regime, T9 proves the *harmful/neutral* regime, T2/T6/T7 govern the *observability* of the evidence Z (and hence the boundary of the *unknowable* regime), T5 supplies the operational certificate, and GDR certifies abstention. This is the fusion-case shadow of the general {adapt, freeze, abstain} theory.